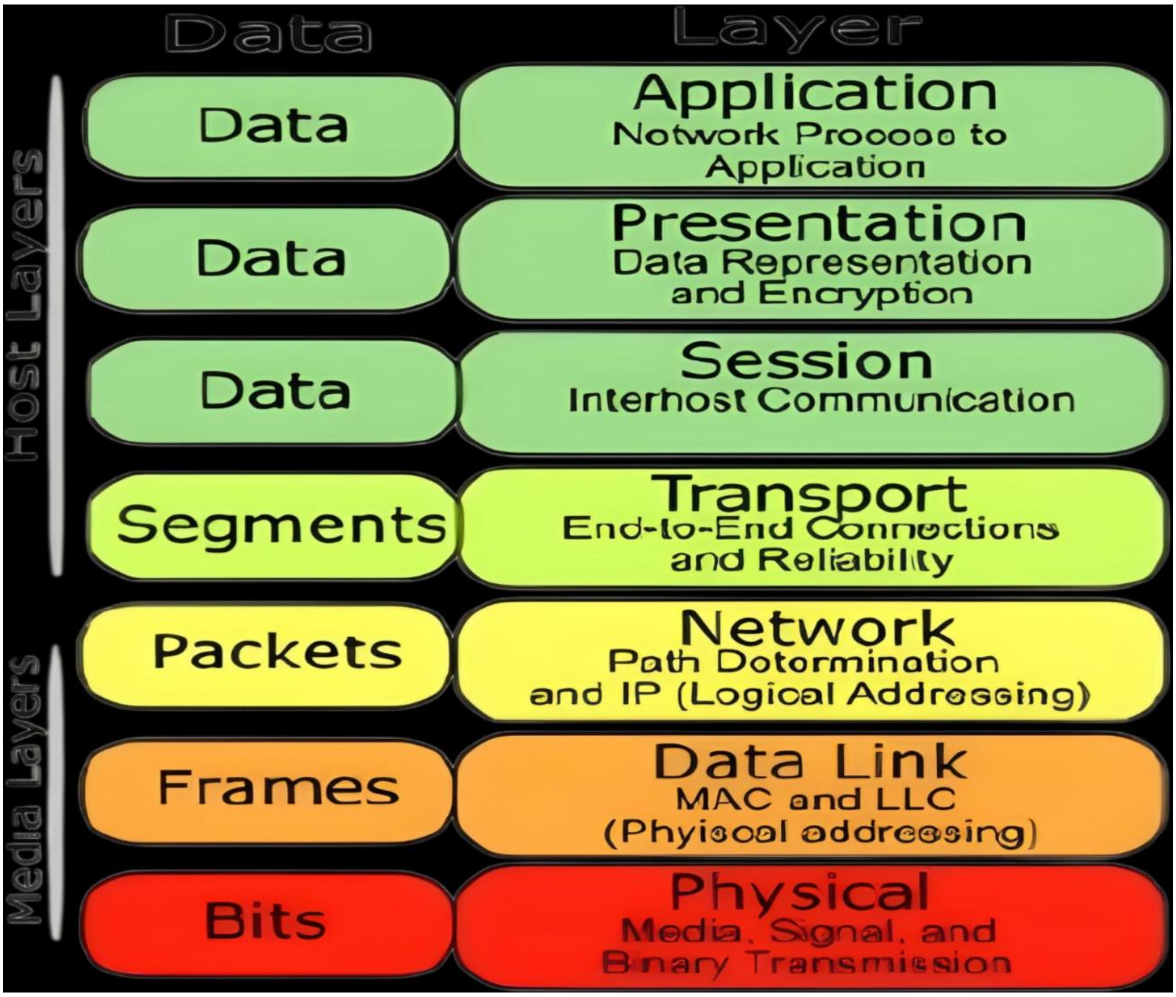
OSI Layer

1. Image of OSI layer :



2. Matching Scenarios to OSI Layers :

- **a) Encrypting data before sending:** Presentation Layer (Layer 6)
- **b) Routing packets across networks:** Network Layer (Layer 3)
- **c) Converting data to electrical signals:** Physical Layer (Layer 1)

- **d) Opening a web page in your browser:** Application Layer (Layer 7)
- **e) Detecting and correcting errors in frames:** Data Link Layer (Layer 2)

3. How OSI Layers Work in a Daily App (WhatsApp) :

When sending a text message or photo on **WhatsApp**, data moves through these key layers:

- **1. Application Layer (Layer 7):** You type a message and hit "Send" in the WhatsApp interface. The app uses protocols like HTTPS/XMPP to package the chat request.
- **2. Presentation Layer (Layer 6):** WhatsApp encrypts your message end-to-end so nobody in between can read it, and compresses any image attachments to save data.
- **3. Transport Layer (Layer 4):** WhatsApp breaks the message into smaller segments and assigns **TCP** (Transmission Control Protocol) to ensure every piece arrives reliably and in the correct order.
- **4. Network Layer (Layer 3):** Attaches your device's IP address and WhatsApp's server IP address to route the packets across the internet to the destination.

4. Troubleshooting Website Connection vs. Successful Ping

Identified Problem Layers: Layer 7 (Application Layer) or Layer 6 (Presentation Layer) :

Reasoning

1. **Layers 1 to 4 are Working correctly:**
    - A successful ping uses the **ICMP** protocol, which operates at **Layer 3 (Network Layer)**.
    - Because the ping gets a response, physical cables/signals (Layer 1), local Wi-Fi/switch frames (Layer 2), IP routing (Layer 3), and basic network connectivity (Layer 4) are completely functional.
2. **The Failure is at the Higher Layers:**
    - **Layer 7 (Application):** The web server software (like Apache or Nginx) on the target server might be down, or the Web Browser itself might be blocking the connection.
    - **Layer 6 (Presentation):** An SSL/TLS security certificate mismatch (HTTPS error) can stop the browser from rendering the site even though the server IP responds.
    - **DNS Resolution (Layer 7):** If pinging the *IP address* works but typing the *domain name* (e.g., google.com) fails, the DNS service is unable to convert the web address into the IP.